King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 223

Original exam date: 31 July 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If $f(x) = 5 - 2x^3$ has an inverse function, then $(f^{-1})'(7) =$

(a) $\frac{1}{6}$

(b) $\frac{1}{5}$

(c) $\frac{1}{3}$

(d) $\frac{-1}{6}$

(e) $\frac{-1}{5}$

2. If the function $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ has a relative minimum at $(0, 0)$ and a relative maximum at $(2, 2)$, then $a_3 + a_2 + a_1 + a_0 =$

(a) 2

(b) 1

(c) 0

(d) -2

(e) -1

3. A ladder 10 meters long is leaning against the wall of a house. The base of the ladder is pulled away from the wall at a rate of $\frac{2}{3}$ meters per second (m/s). Consider the triangle formed by the side of the house, the ladder, and the ground. The rate (in m^2/s) at which the area of the triangle is changing when the base of the ladder is 6 meters from the wall is

(a) $\frac{5}{6}$

(b) 1

(c) $\frac{7}{6}$

(d) $\frac{5}{2}$

(e) $\frac{3}{2}$

4. The sum of all values of a in the interval $(0, 2\pi)$ for which the function $f(x) = \cos^2(2x)$ has a relative extremum at $x = a$ is

(a) 4π

(b) 8π

(c) 5π

(d) 7π

(e) 6π

5. If $y = \dfrac{(2x+1)^3(x^2-1)^2}{x+3}$, then $y'(0) =$

(a) $\dfrac{16}{9}$

(b) $\dfrac{17}{9}$

(c) 2

(d) $\dfrac{5}{3}$

(e) $\dfrac{19}{9}$

6. The value of c which satisfies the conclusion of the Mean Value Theorem for the function $f(x) = (x+3)\ln(x+3)$ on $[-2, -1]$ is

(a) $3 - 3e$

(b) $4 - 3e$

(c) $\dfrac{2-3e}{e}$

(d) $\dfrac{3-3e}{e}$

(e) $\dfrac{4-3e}{e}$

7. If $f(t) = \dfrac{3^{2t}}{t}$, then $f'(1) =$

(a) $9\ln 3 - 1$

(b) $18\ln 3 - 9$

(c) $18\ln 3$

(d) $9\ln 3 - 2$

(e) $18\ln 3 + 4$

8. The value of c which satisfies the conclusion of Rolle's theorem for the function $f(x) = (x^2 - 2x)\, e^x$ on $[0, 2]$ is

(a) $\dfrac{\sqrt{2}}{2}$

(b) $\dfrac{1}{2}$

(c) 1

(d) $\sqrt{2}$

(e) $-\sqrt{2}$

9. Let $f(x) = x^3 - \dfrac{3}{2}x^2$ on $[-1, 2]$ and let M be the absolute maximum of $f(x)$ and m be the absolute minimum of $f(x)$. Then $M + m =$

(a) 1

(b) $\dfrac{3}{2}$

(c) $\dfrac{-3}{4}$

(d) $\dfrac{-1}{2}$

(e) $\dfrac{-5}{2}$

10. By using Newton's method to approximate a zero of the function $f(x) = x^3 + x - 1$ using the initial guess $x_1 = \dfrac{1}{2}$, we get $x_2 =$

(a) $\dfrac{3}{7}$

(b) $\dfrac{5}{7}$

(c) $\dfrac{6}{7}$

(d) $\dfrac{4}{7}$

(e) 1

11. If $g(x) = \left(\dfrac{3x^2 - 2}{2x + 3}\right)^{-2}$, then $g'(0) =$

(a) 3

(b) 4

(c) 5

(d) 2

(e) 1

12. If $x^2 + y^2 = 25$, then $\dfrac{d^2y}{dx^2} =$

(a) $\dfrac{-5}{y^3}$

(b) $\dfrac{-25}{y^3}$

(c) $\dfrac{-25}{y^2}$

(d) $\dfrac{25}{y^3}$

(e) $\dfrac{5}{y^3}$

13. The sum of all critical numbers of the function $f(t) = t\sqrt{4-t}$ in the interval $(-\infty, 3)$ is

(a) $\dfrac{8}{3}$

(b) $\dfrac{7}{3}$

(c) $\dfrac{10}{3}$

(d) 3

(e) 2

14. The normal line to the parabola $y = x^2$ at the point $(2, 4)$ intersects the parabola a second time at

(a) $\left(\dfrac{-2}{3}, \dfrac{4}{9}\right)$

(b) $\left(\dfrac{-7}{4}, \dfrac{49}{16}\right)$

(c) $(-1, 1)$

(d) $\left(\dfrac{-9}{4}, \dfrac{81}{16}\right)$

(e) $(4, 16)$

15. The function $f(x) = \dfrac{x^2}{2x-1}$ is decreasing on the interval

(a) $\left(\dfrac{1}{2}, 2\right)$

(b) $(-\infty, 0)$

(c) $(1, \infty)$

(d) $\left(-1, \dfrac{1}{2}\right)$

(e) $\left(0, \dfrac{1}{2}\right)$

16. All edges of a cube are expanding at a rate of 6 centimeters per second (cm/s). How fast (in cm^3/s) is the volume changing when each edge is $2\,cm$?

(a) 108

(b) 36

(c) 72

(d) 60

(e) 144

17. The number of points at which the graph of the equation $4x^2 + y^2 - 8x + 4y + 4 = 0$ has a vertical tangent line is

(a) 1

(b) 2

(c) 4

(d) 0

(e) 3

18. The slope of the graph of the equation $\arcsin(xy) = \dfrac{2}{3}\arctan(2x)$ at the point $\left(\dfrac{1}{2}, 1\right)$ is

(a) $\dfrac{2\sqrt{3}-6}{5}$

(b) $\dfrac{2\sqrt{3}+5}{3}$

(c) $\dfrac{2\sqrt{3}-6}{3}$

(d) $\dfrac{2\sqrt{3}+4}{3}$

(e) $\dfrac{\sqrt{3}-4}{2}$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$f(-1)=7$, $f'(-1)=-6$, so $(f^{-1})'(7)=-\frac{1}{6}$.
2	**(b)**	$a_1=a_0=0$, $a_2=-3a_3$, $-4a_3=2$: $a_3=-\frac{1}{2}$, $a_2=\frac{3}{2}$, sum $=1$.
3	**(c)**	$y=8$, $\dot{y}=-\frac{1}{2}$; $\dot{A}=\frac{1}{2}(\frac{2}{3}\cdot 8-3)=\frac{7}{6}$.
4	**(d)**	$f'=-2\sin 4x=0$: $x=\frac{k\pi}{4}$, $k=1..7$, all sign changes; sum $=7\pi$.
5	**(b)**	Log differentiation: $y(0)=\frac{1}{3}$, $\frac{y'}{y}=6-\frac{1}{3}$, so $y'(0)=\frac{17}{9}$.
6	**(e)**	Slope $2\ln 2=\ln(c+3)+1$, so $c+3=\frac{4}{e}$ and $c=\frac{4-3e}{e}$.
7	**(b)**	Quotient rule: $f'(1)=2\ln 3\cdot 9-9=18\ln 3-9$.
8	**(d)**	$f'=(x^2-2)e^x=0$, so $c=\sqrt{2}$ (the one in $(0,2)$).
9	**(d)**	Values $-\frac{5}{2},0,-\frac{1}{2},2$: $M=2$, $m=-\frac{5}{2}$, sum $-\frac{1}{2}$.
10	**(b)**	$x_2=\frac{1}{2}-\frac{-3/8}{7/4}=\frac{1}{2}+\frac{3}{14}=\frac{5}{7}$.
11	**(a)**	$g'=-2h^{-3}h'$ with $h(0)=-\frac{2}{3}$, $h'(0)=\frac{4}{9}$: $g'(0)=3$.
12	**(b)**	$y'=-\frac{x}{y}$, $y''=-\frac{x^2+y^2}{y^3}=-\frac{25}{y^3}$.
13	**(a)**	$f'=\frac{8-3t}{2\sqrt{4-t}}=0$ at $t=\frac{8}{3}$ (and $t=4$ is outside), so the sum is $\frac{8}{3}$.
14	**(d)**	Normal: $y=-\frac{x}{4}+\frac{9}{2}$; $4x^2+x-18=0$ gives $x=-\frac{9}{4}$, point $(-\frac{9}{4},\frac{81}{16})$.
15	**(e)**	$f'=\frac{2x(x-1)}{(2x-1)^2}<0$ on $(0,\frac{1}{2})\cup(\frac{1}{2},1)$.
16	**(c)**	$V=s^3$: $V'=3s^2s'=3(4)(6)=72$.
17	**(b)**	$y'=\frac{4(1-x)}{y+2}$; $y=-2$ gives $x=0,2$: 2 points.
18	**(c)**	Implicit: $\frac{2}{\sqrt{3}}(1+\frac{y'}{2})=\frac{2}{3}$, so $y'=\frac{2\sqrt{3}-6}{3}$.